Module 2 Challenge

Calculus Honors · Limits Without Bound & Special Limits · 16 points

Name: _____ Date: _____

> This is your fullest, most careful work of the module. Show every step in clean vertical layout, justify your reasoning in words where asked, and label your final answers. You will be graded on understanding, quality of reasoning, clarity and precision, and intellectual honesty (say clearly where you are certain and where you are not).

1. End-behavior analysis (6 pts). Consider $f(x) = \dfrac{2x^2 - 3x}{x^2 + 4}$. (a) Find $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$, and state the horizontal asymptote(s). (b) Determine whether f has any vertical asymptotes, and justify your answer by examining the denominator.

2. A Squeeze proof (5 pts). Prove, using the Squeeze Theorem, that $\lim_{x \to 0} x^4 \sin(1/x^2) = 0$. State your bounding inequality explicitly, evaluate both outer limits, and name the theorem's hypothesis that makes your conclusion valid.

3. Classify and resolve (5 pts). For each limit, state which indeterminate form it presents (if any) and evaluate it, showing the technique you used. (a) $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$ (b) $\lim_{x \to \infty} \dfrac{x^2 - 1}{3x^2 + 2x}$ (c) $\lim_{x \to 0} \dfrac{\sin x}{x + x^2}$. For part (c), note clearly which technique or special limit from this module you needed, and how confident you are in each of your three answers.